

Aditya Kumar

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SUMMARY

MS student in Computational Fluid Dynamics and Aeronautics; builds and verifies GPU-accelerated solvers and physics-informed surrogate models for fluid and thermal systems using CUDA, MPI, Python, and OpenFOAM.

EDUCATION

University of Washington

MS Aeronautics and Astronautics (Fluids Concentration), GPA: 3.88/4.0

Seattle, WA

Sep 2025 - Present

Vishwakarma Institute of Technology

BTech Mechanical Engineering, CGPA: 8.42/10 (Top 10%)

Pune, India

Jul 2021 - May 2025

EXPERIENCE

Research Assistant, Computational Fluid Mechanics Lab

University of Washington

Seattle, WA

Sep 2025 - Present

- Developing a GPU-accelerated Poisson solver (CUDA, Fortran, MPI) on an HPC cluster, benchmarking accuracy against Taylor-Green vortex decay test cases with automated Python post-processing pipelines
- Implementing RANS/LES turbulence models via finite volume discretization with convergence analysis; documenting methodology and presenting results to the research group

Research Co-op, Thermal Systems Engineering

Kirloskar Pneumatic Co. Ltd.

Pune, India

Jul 2024 - May 2025

- Trained a Gaussian Process surrogate (POD dimensionality reduction, Matern kernel) on 800 to 1200 CFD cases of a twin-screw compressor; applied ROM techniques to improve compression efficiency by 2%
- Conducted DOE studies and automated numerical workflows for thermodynamic cycle prediction, integrating data visualization pipelines with Matplotlib and Pandas

Aerodynamics Engineer

Veloce Racing India (Formula Student)

Pune, India

Aug 2021 - Apr 2023

- Automated multi-physics CFD workflows in ANSYS Fluent and STAR-CCM+, achieving 5% drag reduction through computational design optimization across 30+ full-body CFD runs
- Coordinated wind tunnel validation campaigns and mentored 5 junior engineers on CFD best practices (team finished 1st nationally, 2nd globally in the combustion category)

Manufacturing Engineering Intern

Jaguar Land Rover

Pune, India

May 2023 - Aug 2023

- Optimized Oil Filling and Cotter Press machinery through process analysis and design-for-manufacturing improvements, reducing unplanned downtime by 15%

PROJECTS

2D Incompressible Navier-Stokes Solver

Julia, CairoMakie

May 2026 - Present

- Built a 2D incompressible Navier-Stokes solver from scratch on a staggered MAC grid with RK3 fractional-step time integration; full test suite passes 15 of 15
- Validated second-order spatial accuracy via a method of manufactured solutions convergence study and matched centerline velocity against Ghia et al. (1982) for lid-driven cavity flow at Re 100 and Re 400

Conjugate Heat Transfer Study, Liquid-Cooled Cold Plate

OpenFOAM, Python, SLURM

Apr 2026 - Aug 2026

- Automated a 96-case conjugate heat transfer design sweep of a liquid-cooled microchannel cold plate under a 700W GPU-class heat load via a templated SLURM array pipeline (89/96 converged)
- Trained Gaussian Process surrogate models and ran constrained optimization, cutting thermal resistance 22% at matched pumping power; validated via GCI study (2.21% on pressure drop)

TECHNICAL SKILLS

Programming: Python, MATLAB, Fortran, C, C++, CUDA, Julia, Bash/Shell, SQL

HPC and Parallel Computing: MPI, OpenMP, GPU Computing (CUDA), SLURM, PBS, Cluster Computing

CFD and Simulation: ANSYS Fluent, PyAnsys (PyFluent), STAR-CCM+, OpenFOAM, ParaView

ML and Data Tools: PyTorch, TensorFlow, Scikit-learn, NumPy, Pandas, Streamlit

Numerical Methods: Finite Volume Discretization, RANS/LES Turbulence Modeling, DOE, ROM